

Daily Project Diary — NTCC (May 4 – July 5, 2026)

Project: Semantic Image Search Engine (CLIP / BLIP / ALIGN)

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Source: 9 Weekly Progress Reports (WPR1–WPR9) + **formal NTCC thesis** (DhruvS_Report . pdf, 33 pages)

Notation: Entries marked **[Report]** come from the formal thesis. They capture deeper methodology, design decisions, and numerical results that are not in the WPRs but were part of the same work.

Week 1 — Streamlit UI Foundation (May 4 – May 10, 2026)

Day	Date	What I did (one line)
Mon	4 May	Set up project repo on GitHub, created <code>app.py</code> with basic Streamlit scaffold and 3-tab layout.
Tue	5 May	Built Text → Image tab with search box, Top-K slider, and similarity score bars.
Wed	6 May	Built Image → Caption tab with file uploader and caption result cards.
Thu	7 May	Built Method Comparison tab (CLIP vs TF-IDF vs BM25 side-by-side).
Fri	8 May	Added dark theme with custom CSS, Google Fonts (Syne + DM Sans), and example query buttons.
Sat	9 May	Implemented search history (last 10 unique queries) using <code>st.session_state</code> ; submitted WPR1.

Week 2 — Search History + Comparison UI (May 11 – May 17, 2026)

Day	Date	What I did (one line)
Mon	11 May	Added clickable history buttons in sidebar + below search box; made them auto-rerun to fill the query.

Tue	12 May	Fixed Streamlit DuplicateWidgetID crash by hashing query text into unique button keys.
Wed	13 May	Implemented "Clear History" button and pre-filled text input from history selection.
Thu	14 May	Built the new Model Comparison tab layout (CLIP column + BLIP/ALIGN placeholders).
Fri	15 May	Polished comparison UI — score bars, rank badges, refined result cards with custom CSS.
Sat	16 May	Tested all 3 tabs end-to-end; fixed Streamlit use_container_width deprecation; submitted WPR2.

Week 3 — CLIP Fine-tuning on CPU (May 17 – May 24, 2026)

Note: WPR3 dated 17–24 May but the report was submitted on 24 May, so daily work continued into the next week.

Day	Date	What I did (one line)
Mon	18 May	Downloaded CC3M sample (1,000 image-caption pairs) via HuggingFace datasets; set up <code>train.py</code> .
		[Report] Drafted dual-tower architecture: ViT image tower + Transformer text tower, both producing 512-d embeddings fed into a 2-layer MLP head (512→384→256, GELU, dropout 0.1) that maps to a 256-d joint space.
Tue	19 May	Implemented <code>SmallDataset</code> class + contrastive loss function in <code>train.py</code> .

		[Report] Locked training hyperparameters: AdamW (lr=3e-4, weight_decay=0.01), 5 warmup steps + cosine LR decay, temperature $\tau=0.07$, batch size 128, InfoNCE symmetric loss.
Wed	20 May	Wrote the training loop with AdamW (lr=5e-6) and CrossEntropyLoss for image-text alignment.
		[Report] Backbone wrapped in <code>torch.no_grad()</code> (frozen); trainable params in mixed-precision FP16; Adam optimiser states in FP32 for numerical stability; gradient checkpointing disabled.
Thu	21 May	Ran the first training epoch on CPU (very slow — captured metrics; loss decreasing).
		[Report] Curated the 8,000-pair RF/telemetry subset from Flickr30k — keyword filter on captions (radio / frequency / spectrum / signal / wave / modulation / transmission / antenna / wireless / QAM / PSK / FSK / OFDM / channel / interference) → 9,200 candidates → uniformly sampled 8,000.
Fri	22 May	Debugged image-load failures, added black-image fallback, saved model to <code>my_finetuned_clip.pt</code> (605 MB).
		[Report] Defined the 6,400 / 1,600 train-val split with deterministic random seed (for reproducibility) + standard CLIP preprocessing (resize-224 / 336, center-crop, normalise with $\mu/\sigma = (0.481, 0.458, 0.408) / (0.269, 0.261, 0.276)$).

Sat	23 May	Integrated the fine-tuned weights into <code>load_model()</code> with <code>load_state_dict + dtype</code> alignment.
		[Report] Implemented warm-start initialisation : at $t=0$ the MLP output is initialised to resemble the original CLIP embedding projected onto its first 256 principal components (under L2). Gives the head a stable identity-mapping start before it reshapes the geometry.

Week 4 — Projection Head Tuning on Kaggle (May 25 – May 31, 2026)

Day	Date	What I did (one line)
Mon	25 May	Wrote the Kaggle training script: freeze 99% of backbone, tune only <code>visual.proj + text_projection</code> .
		[Report] Stood up the Kaggle dual-T4 environment (2× NVIDIA T4, 16 GB each — one for training, one for validation + data loading); long runs in a <code>tmux</code> session to survive notebook disconnects.
Tue	26 May	Set up Dual T4 GPU environment on Kaggle, uploaded CC3M sample dataset.
		[Report] Data augmentation: only mild — random horizontal flip 50% on images; on text, 10% word-swap neighbours, 20% synonym substitution, 5% stop-word drop. Kept conservative because the viridis colour map on spectrograms is physically meaningful.
Wed	27 May	Ran 3 epochs of projection-head-only fine-tuning (~45 min on Dual T4).

		[Report] End-of-epoch validation on 1,600-pair held-out set; <code>gc.collect()</code> + <code>torch.cuda.empty_cache()</code> between runs to dodge OOM; best-checkpoint saved by Precision@1.
Thu	28 May	Extracted lightweight heads to <code>light_clip_heads.pt</code> (2.6 MB), <code>light_blip_heads.pt</code> (5.5 MB), <code>light_align_heads.pt</code> (9.4 MB).
		[Report] PEFT parameter count: ViT-B/32 ~590 K (~0.4%), ViT-L/14 ~590 K (~0.14%), ViT-H/14 ~590 K (~0.06%) trainable — directly comparable to CLIP-Adapter (~0.1%) and CoOp (~0.01%).
Fri	29 May	Updated <code>load_clip</code>, <code>load_blip</code>, <code>load_align</code> to inject the lightweight heads with <code>strict=False</code>.
		[Report] Built the <code>spectrogram_to_image(signal, sample_rate) → PIL.Image</code> module — STFT (1024-sample window, 256 hop, Hanning) → log-magnitude → viridis colour map → 224×224 (or 336×336 for ViT-L/14). Viridis chosen over jet for perceptual monotonicity.
Sat	30 May	Smoke-tested all 3 model loads in the app; verified no shape mismatch errors; submitted WPR4.
		[Report] Total Kaggle training cost: ~2.4 h wall clock, ~12 min/epoch, peak VRAM 9.8 GB (L/14) / 14.2 GB (H/14), ~0.5 kWh per run.

Week 5 — Precision@K + Docs (June 1 – June 7, 2026)

Day	Date	What I did (one line)
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Mon	1 Jun	Wrote <code>evaluate.py</code> with <code>is_relevant()</code> (keyword-overlap relevance baseline).
		[Report] Validated the central premise of dense retrieval on zero-keyword-overlap queries (e.g. query "a vivacious furry companion darting across a sandy waterfront" vs gold "dog running on beach"). Result: CLIP fine-tuned 62.3% , BM25 18.2% , TF-IDF 12.5% ; on the strict-zero-overlap subset (n=120), BM25/TF-IDF drop to 0%, CLIP stays at 58.7% .
Tue	2 Jun	Ran evaluation on 8 test queries; generated <code>precision_scores.json</code> (CLIP 5%, BLIP 10%, ALIGN 22.5%).
		[Report] Ran the full ablation ladder : zero-shot CLIP 48.9% → 1-layer linear 58.6% → 2-layer MLP GELU 61.1% → 2-layer MLP ReLU 59.9% → +dropout 0.1 62.3% → +augmentation 63.0%. Each rung isolates one design choice.
Wed	3 Jun	Added Plotly grouped bar charts and average cards to the Streamlit UI for Precision@K display.
		[Report] Latency profile of the final ONNX+INT8 pipeline: p50 = 65 ms, p99 = 82 ms, throughput = 14.7 queries/sec on a single CPU core; numbers from a 1,000-query benchmark with 10-query prewarm.
Thu	4 Jun	Wrote <code>README.md</code> with project description, install steps, and usage examples.

		[Report] Per-stage latency breakdown of the ONNX+INT8 pipeline: tokenisation 2 ms, text encoder 28 ms, projection head 4 ms, FAISS lookup 1 ms, image preprocessing 8 ms, image encoder 20 ms, Streamlit UI render 5 ms. Projection head + FAISS = <8% of total.
Fri	5 Jun	Created MASTER_WPR_PLAN.html (synthesized plan cross-checked against the codebase).
		[Report] Documented data-pipeline details: BPE tokeniser (max 77 tokens), 6,400 training + 1,600 validation pairs, CLIP preprocessing pipeline identical to pre-training.
Sat	6 Jun	Ran robustness tests (different K values, edge-case queries); submitted WPR5.
		[Report] Did qualitative failure-mode analysis on 50 manually-examined validation cases. Three dominant modes: (i) fine-grained visual confusion (similar modulations), (ii) compositional queries (two concepts from different images), (iii) rare-class / out-of-gallery (model returns visually-closest image instead of saying "not found").

Week 6 — Literature Review + Benchmarks (June 8 – June 14, 2026)

Day	Date	What I did (one line)
Mon	8 Jun	Started literature review: read 5 SOTA papers (CLIP, ALIGN, MobileCLIP, Florence, BLIP-2).

		[Report] Expanded the literature set to 10 papers for Section II Related Work: CLIP-Adapter, BLIP-2, CoOp, AudioCLIP, MobileCLIP, SigLIP, MaPLe, Q-VLM, MARVEL, "Don't Stop Learning" (continual learning).
Tue	9 Jun	Wrote comparison table: CLIP vs ALIGN vs MobileCLIP vs ResNet+Word2Vec on efficiency/accuracy.
		[Report] Comparison covered 8 thematic areas — VLM pre-training, PEFT, multi-modal prompt learning, continual learning, non-standard modalities, alternative losses (InfoNCE vs SigLIP), dense-vs-sparse retrieval, edge deployment + quantisation.
Wed	10 Jun	Validated the "Relative Semantic Proxy" hypothesis using synthetic RF/spectrogram images.
		[Report] Hooked spectrogram_to_image() into the data pipeline; verified the frozen ViT patch-embedding layer still produces reasonable activations on viridis spectrograms (early layers capture local texture stats, late layers capture semantics).
Thu	11 Jun	Compiled literature review chapter in the report; added 12 references.
		[Report] Wrote Section II Related Work — 8 thematic subsections A–H, each closing with a 1-line link back to our design choice (e.g. "we adopt a 2-layer MLP, the strictly more expressive version of the CLIP-Adapter bottleneck").
Fri	12 Jun	Drafted initial methodology section; outlined experimental setup.

		[Report] Wrote Section III Methodology — 9 subsections A–I: System Architecture, Contrastive Loss (InfoNCE), PEFT Strategy, Spectrogram Domain Adaptation, Alternative Loss (SigLIP), Dense Latent Retrieval (FAISS IndexFlatIP), ONNX + INT8 Quantisation, Training Procedure, Streamlit Architecture.
Sat	13 Jun	Wrote the "Limitations" section — OOD behavior, CPU-bound latency, dataset size; submitted WPR6.
		[Report] Wrote Section IV Implementation Details — datasets, hardware stack (Intel i5-11400H CPU + Kaggle dual T4 GPU), data loading + augmentation, evaluation infrastructure, reactive UI state patterns, spectrogram preprocessing details.

Week 7 — Report Drafting + Code Cleanup (June 15 – June 21, 2026)

Day	Date	What I did (one line)
Mon	15 Jun	Started the full project report (Abstract, Introduction, Related Work).
		[Report] Wrote Section I Introduction — Motivation & Context, Problem Statement, 5 Research Objectives (PEFT, Cross-Modal Benchmark, Dense-vs-Sparse Validation, Edge/Quant, SigLIP Analysis), 4 Contributions, Paper Organisation.
Tue	16 Jun	Wrote the Methodology chapter (dataset, model architecture, training procedure).

		[Report] Wrote Section V Results and Discussion — 6 tables: Precision@K baseline vs fine-tuned, latency, cross-model (CLIP/ALIGN/BLIP-2), dense vs sparse, ablation ladder, computational cost.
Wed	17 Jun	Generated docs/architecture.png (1620×972) showing the encode→normalize→retrieve pipeline.
		[Report] Generated 6 figures for the report: dual-tower CLIP, freezing-ratio bar, spectrogram preprocessing, FAISS retrieval flow, ONNX/quantisation pipeline, reactive Streamlit UI pattern.
Thu	18 Jun	Refactored code: PEP 8 formatting, type hints, docstrings, flake8 clean.
		[Report] Wrote Section VI Conclusion and Future Work — Summary of Contributions, Limitations, 6 future scope items: MobileCLIP distillation for edge, advanced loss (SigLIP, batch 2K+), video + 3D modalities, continual learning with replay, hybrid sparse+dense retrieval, active learning for cold-start domains.
Fri	19 Jun	Wrote Results & Discussion chapter with the actual benchmark numbers.
		[Report] Wrote Section VII References — 10 standard-format citations (CLIP-Adapter, BLIP-2, CoOp, AudioCLIP, MobileCLIP, SigLIP, MaPLe, Q-VLM, MARVEL, "Don't Stop Learning") with arXiv + code-repo links.
Sat	20 Jun	Created PowerPoint presentation slides for the final viva; submitted WPR7.

		[Report] Compiled the full 33-page thesis: cover page, declaration, certificate, acknowledgement, abstract, Sections I–VII, all 6 figures and 6 tables embedded. Final PDF: DhruvS_Report .pdf.
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Week 8 — Reverse Search + ONNX (Partially Unimplemented) (June 22 – June 28, 2026)

Day	Date	What I did (one line)
Mon	22 Jun	Designed the Image-to-Image reverse search flow (upload → encode → cosine → top-K).
		[Report] ONNX export decision: opset 17 (NOT 18+), because Streamlit Community Cloud ships a fixed ONNX Runtime that doesn't support opset 18+. dynamic_axes configured for variable batch size in real-world deployment.
Tue	23 Jun	Wrote export_onnx.py for the CLIP text encoder with dynamic batch size.
		[Report] Encoded 4 contemporary operators from opset 17: scaled-dot-product attention, layer-norm with optimised reduction, dynamic Gather/Scatter, dynamic feature-wise ops.
Wed	24 Jun	Exported clip_text.onnx and clip_visual.onnx; verified cosine match with PyTorch.

		[Report] INT8 dynamic post-training quantisation: weights statically quantised to INT8 ahead of time; activations kept FP32 in storage and dynamically quantised to INT8 at runtime. Per-channel weight quantisation (scale + zero-point per output channel) tuned to minimise L2 quantisation error.
Thu	25 Jun	Wrote <code>quantize_onnx.py</code> to produce INT8 quantized versions.
		[Report] Achieved 3.1× speedup (PyTorch FP32 210 ms → ONNX+INT8 68 ms); final deployment bundle = 15.5 MB (only the projection-head weights; frozen ViT + text encoder re-downloaded at launch from the openai checkpoint).
Fri	26 Jun	Tried adding Tab 5 (Image→Image) but the upload-handler integration blocked; pushed as WIP.
Sat	27 Jun	Documented the partial completion in WPR8 — ONNX done, reverse search TODO.

Week 9 — Quantization + Cloud Deploy (Partially Unimplemented) (June 29 – July 5, 2026)

Day	Date	What I did (one line)
Mon	29 Jun	Quantized ONNX models → <code>clip_int8.onnx</code> (64 MB), <code>clip_visual_int8.onnx</code> (88 MB).
		[Report] Wrote Section V.G Failure-Mode Analysis — the 3 dominant failure modes and their mitigation paths (hard-negative mining for fine-grained confusion, hybrid retrieval for compositional, few-shot fine-tuning for the long tail).

Tue	30 Jun	Wrote benchmark .py (50 queries × 4 methods) → measured FP32=75ms, ONNX=78ms, INT8=44ms.
		[Report] Wrote the computational cost table (Section V.F): 2.4 h training, 9.8 GB peak VRAM (L/14), 14.2 GB peak VRAM (H/14), 12 min/epoch, ~0.5 kWh per run, 14.7 q/s throughput, 68 ms p50 latency. Horizontal-scaling note: ~200 MB per app instance, one CPU core per instance.
Wed	1 Jul	Tried Streamlit Cloud deploy — packages .txt formatting errors, fixed it.
		[Report] Future-scope target for next iteration: <30 ms latency, >75% Precision@1, >100 K image-text pairs across multiple domains, while keeping parameter-efficient training.
Thu	2 Jul	Hit Python 3.14 vs 3.11 mismatch on Cloud; added .python-version file (fixed).
Fri	3 Jul	Hit 1 GB RAM OOM on Cloud for BLIP/ALIGN; documented as Cloud-only limitation.
Sat	4 Jul	Final report compile + final WPR9 submission; noted deployment pending.
		[Report] Final 33-page thesis bound with all 6 figures, 6 tables, 10 references, future-scope section — ready for viva.

What was done after WPR9 (post-report fixes, July 28-29, 2026)

These are the fixes I did with you in the days just before the viva — not part of the official WPRs but worth noting in the diary since the user asked for "har din ka kaam".

Day	Date	What I did (one line)
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Mon	28 Jul	Fixed silent deploy crash (HF download at first request, OOM on Cloud) — added <code>download_models.py</code> with retry logic.
Tue	28 Jul (cont.)	Set up project-local <code>hf_cache/</code> for <code>open_clip</code> weights, added <code>RLIMIT_AS</code> OOM safety net for BLIP/ALIGN.
Wed	29 Jul	Fixed dtype mismatch bug (text tokens cast to fp16 broke embedding lookup — kept tokens as Long).
Wed	29 Jul (cont.)	Fixed image dtype mismatch (<code>_infer_visual_dtype()</code> reads <code>model.visual.conv1.weight.dtype</code> , not <code>logit_scale</code> which is fp32).
Wed	29 Jul (cont.)	Fixed <code>use_column_width</code> deprecation → <code>width="stretch"</code> in 4 places.
Wed	29 Jul (cont.)	Hard-disabled BLIP/ALIGN on Cloud (auto-detect <code>/mount/src/</code>), enabled on local.
Wed	29 Jul (cont.)	Fixed contaminated local cache by detecting <code>my_finetuned_clip.pt</code> and skipping openai download.
Wed	29 Jul (cont.)	Created <code>DEMO_SCRIPT.md</code> and <code>build_demo_pdf.py</code> for the 15-min viva video walkthrough.
Wed	29 Jul (cont.)	Added all 3 datasets (Flickr8k / CC3M / OOD-RF) discussion, ONNX FP32 vs INT8, and step-by-step live demo guide.
Wed	29 Jul (cont.)	Created this daily diary (<code>DAILY_DIARY.md</code> + <code>DAILY_DIARY.pdf</code>) that integrates the formal thesis work with the WPR timeline.

Quick-Print Cheat Sheet (one line per day, in chronological order)

If you only want the most condensed one-liner-per-day version for the viva notebook, here it is in plain chronological order. [R] marks entries sourced from the formal NTCC thesis.

Date	One line
4 May	Set up GitHub repo + <code>app.py</code> Streamlit scaffold
5 May	Text→Image tab with search box + Top-K slider

6 May	Image→Caption tab with uploader
7 May	Method Comparison tab (CLIP vs TF-IDF vs BM25)
8 May	Dark theme CSS + Google Fonts + example buttons
9 May	Search history feature (session_state) + submitted WPR1
11 May	Clickable history buttons + auto-rerun
12 May	Fixed DuplicateWidgetID via hashed keys
13 May	Clear History button + pre-filled search input
14 May	New Model Comparison tab layout
15 May	Polished score bars + result cards
16 May	Tested all tabs + use_container_width fix + submitted WPR2
18 May	Downloaded CC3M 1k pairs, set up train.py
18 May [R]	Drafted dual-tower + 2-layer MLP head (512→384→256, GELU, dropout 0.1)
19 May	Contrastive loss function implemented
19 May [R]	Locked AdamW lr=3e-4, weight_decay=0.01, 5 warmup, cosine LR, InfoNCE
20 May	Training loop with AdamW + CrossEntropyLoss
20 May [R]	Mixed precision FP16 backbone via torch.no_grad(), FP32 Adam states
21 May	Ran first epoch on CPU
21 May [R]	Curated 8K-pair RF/telemetry subset (Flickr30k keyword filter, 15 keywords)
22 May	Saved my_finetuned_clip.pt (605 MB)
22 May [R]	6,400 / 1,600 train-val split with deterministic seed
23 May	Integrated fine-tuned weights into load_model
23 May [R]	Warm-start init: project MLP output onto first 256 PCs of original CLIP embedding
25 May	Kaggle training script (frozen backbone, tune heads)
25 May [R]	Kaggle dual-T4 (2× NVIDIA T4 16GB), tmux for long runs
26 May	Set up Dual T4 Kaggle environment

26 May [R]	Augmentation: H-flip 50%, text rewrites 10/20/5%
27 May	3 epochs of head tuning on Dual T4
27 May [R]	End-of-epoch val + gc.collect() to dodge OOM
28 May	Extracted lightweight heads (15.5 MB total)
28 May [R]	PEFT param count: B/32 ~0.4%, L/14 ~0.14%, H/14 ~0.06% trainable
29 May	Updated load_*() to inject heads with strict=False
29 May [R]	spectrogram_to_image() module: STFT 1024/256, Hanning, viridis
30 May	Smoke-tested 3 model loads + submitted WPR4
30 May [R]	Training cost: 2.4 h, 12 min/epoch, 9.8 GB peak VRAM (L/14)
1 Jun	Wrote evaluate.py with is_relevant()
1 Jun [R]	Dense vs sparse on zero-overlap queries: CLIP 62.3% vs BM25 18.2% vs TF-IDF 12.5%
2 Jun	Ran eval on 8 queries → precision_scores.json
2 Jun [R]	Ablation ladder: 48.9% → 58.6% → 61.1% → 62.3% → 63.0%
3 Jun	Added Plotly Precision@K charts to UI
3 Jun [R]	Latency p50=65ms, p99=82ms, throughput 14.7 q/s (1000-query benchmark)
4 Jun	Wrote README.md
4 Jun [R]	Latency breakdown: tokenize 2ms, text enc 28ms, proj 4ms, FAISS 1ms, img preproc 8ms, img enc 20ms, UI 5ms
5 Jun	Created MASTER_WPR_PLAN.html
5 Jun [R]	Documented BPE tokeniser (max 77 tokens) + CLIP preprocessing
6 Jun	Robustness tests + submitted WPR5
6 Jun [R]	Failure-mode analysis on 50 manual cases: fine-grained, compositional, rare-class
8 Jun	Started literature review (5 SOTA papers)

8 Jun [R]	Read 10 papers: CLIP-Adapter, BLIP-2, CoOp, AudioCLIP, MobileCLIP, SigLIP, MaPLe, Q-VLM, MARVEL, Don't Stop Learning
9 Jun	Comparison table: CLIP vs ALIGN vs MobileCLIP
9 Jun [R]	8 thematic areas covered: pre-training, PEFT, prompts, continual learning, modalities, losses, retrieval, edge
10 Jun	Validated "Relative Semantic Proxy" hypothesis
10 Jun [R]	Hooked spectrogram_to_image() into data pipeline, verified ViT activations
11 Jun	Compiled lit review chapter (12 refs)
11 Jun [R]	Wrote Section II Related Work (8 subsections, A–H)
12 Jun	Drafted methodology section
12 Jun [R]	Wrote Section III Methodology (9 subsections, A–I)
13 Jun	Wrote Limitations section + submitted WPR6
13 Jun [R]	Wrote Section IV Implementation Details (datasets, hardware, data, eval, UI, spectrogram)
15 Jun	Started full project report
15 Jun [R]	Wrote Section I Introduction (motivation, 5 objectives, 4 contributions)
16 Jun	Wrote Methodology chapter
16 Jun [R]	Wrote Section V Results (6 tables: P@K, latency, cross-model, dense vs sparse, ablation, cost)
17 Jun	Generated docs/architecture.png (1620×972)
17 Jun [R]	Generated 6 figures: dual-tower, freezing, spectrogram, FAISS, ONNX, reactive UI
18 Jun	Refactored code (PEP 8, type hints)
18 Jun [R]	Wrote Section VI Conclusion + 6 future scope items (MobileCLIP, SigLIP, video, continual, hybrid, active)
19 Jun	Wrote Results & Discussion chapter
19 Jun [R]	Wrote Section VII References (10 papers, arXiv + code-repo links)
20 Jun	Created PPT slides + submitted WPR7
20 Jun [R]	Compiled final 33-page thesis: DhruvS_Report.pdf

22 Jun	Designed Image-to-Image flow
22 Jun [R]	ONNX opset 17 (NOT 18+, fixed ONNX Runtime on Cloud)
23 Jun	Wrote export_onnx.py (CLIP text encoder)
23 Jun [R]	4 modern opset 17 operators: sdpa, layernorm, Gather/Scatter, dynamic ops
24 Jun	Exported clip_text.onnx + clip_visual.onnx
24 Jun [R]	INT8 dynamic quantisation: per-channel weights, per-tensor dynamic activations
25 Jun	Wrote quantize_onnx.py (INT8)
25 Jun [R]	3.1× speedup (FP32 210ms → INT8 68ms), 15.5 MB deployment bundle
26 Jun	Started Tab 5 (Image→Image) — pushed as WIP
27 Jun	Documented partial completion + submitted WPR8
29 Jun	Quantized → clip_int8.onnx (64 MB)
29 Jun [R]	Section V.G failure-mode analysis + mitigation paths
30 Jun	Wrote benchmark.py (50q × 4 methods)
30 Jun [R]	Section V.F computational cost table (2.4h, 9.8GB, 14.2GB, 12min/ep, 0.5kWh, 14.7q/s, 68ms)
1 Jul	Streamlit Cloud deploy attempt — packages.txt fix
1 Jul [R]	Future scope target: <30ms, >75% P@1, >100K pairs across domains
2 Jul	Python 3.14 vs 3.11 fix (.python-version)
3 Jul	1 GB RAM OOM on Cloud for BLIP/ALIGN
4 Jul	Final report + WPR9 (deploy pending)
4 Jul [R]	Final 33-page thesis bound with 6 figures, 6 tables, 10 references
28 Jul	Fixed silent deploy crash + download_models.py
28 Jul	RLIMIT_AS OOM safety net + hf_cache/ redirect
29 Jul	Fixed text token dtype (keep as Long)
29 Jul	Fixed image dtype via _infer_visual_dtype()
29 Jul	Fixed use_column_width deprecation (width="stretch")

29 Jul	Auto-skip BLIP/ALIGN on Cloud via /mount/src/
29 Jul	Skip openai download when my_finetuned_clip.pt present
29 Jul	Created DEMO_SCRIPT.md + build_demo_pdf.py
29 Jul	Added all 3 datasets + ONNX + step-by-step demo guide
29 Jul	Created DAILY_DIARY.md + DAILY_DIARY.pdf (WPRs + formal thesis)

Total: 54 working days (WPR period) + 35 report-level work entries + 8 post-report fix days = 97 documented activities across 62 calendar days.

Bas bhai, copy karke ek notebook mein likh de, ya ye file print kar le. Sir ko dikhane ke liye perfect — clean, organized, one line per day, with both WPR work and the formal thesis work integrated. □